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BANGLADESH

VLSI Design Lab Report

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Abstract

This report presents a comprehensive study of CMOS logic design, sequential circuits, adder circuits, flip-flop implementation, and stick diagram design using DSCH2 and MICROWIND2 tools in the VLSI Design Laboratory. The experiments were conducted to understand the principles of CMOS logic implementation, schematic-level circuit design, timing analysis, and physical layout representation in VLSI systems.

In the initial experiments, basic CMOS logic gates such as NOT, NAND, NOR, AND, OR, and XOR were designed and verified using DSCH2 through truth tables and simulation outputs. Various Boolean expressions were then implemented using CMOS pull-up and pull-down network structures to analyze logic functionality and timing behavior. Sequential circuits including D Flip-Flops and combinational circuits such as Half Adders were also implemented to observe clock-triggered operation, data storage behavior, and binary addition functionality.

Furthermore, stick diagrams of CMOS circuits were implemented in MICROWIND2 to develop an understanding of physical layout representation using polysilicon, diffusion, and metal layers. Stick diagram implementations of Inverter, NAND, NOR, and XOR gates were analyzed through analog simulation outputs to verify correct CMOS operation.

The experiments successfully demonstrated the relationship between Boolean logic, transistor level CMOS implementation, schematic design, timing analysis, and physical layout representation. The knowledge gained from these experiments provides a fundamental understanding of VLSI circuit design methodologies and layout optimization techniques.

Software Information

1. DSCH2

DSCH2 is a digital schematic editor and simulation tool widely used for designing and verifying CMOS-based digital circuits. It allows users to create logic gate schematics, generate truth tables, and analyze timing waveforms through simulation. In this laboratory course, DSCH2 was used to implement combinational and sequential circuits such as logic gates, Boolean functions, adder circuits, and flip-flops.

Features of DSCH2

- Schematic-based CMOS circuit design
- Digital logic simulation and waveform generation
- Timing diagram analysis
- Support for combinational and sequential circuit implementation
- Easy integration with MICROWIND2

2. MICROWIND2

MICROWIND2 is a VLSI layout design and simulation software used to create and analyze CMOS stick diagrams and physical layouts. It provides a graphical environment for designing transistor-level layouts using layers such as polysilicon, diffusion, and metal interconnections. In this laboratory course, MICROWIND2 was used to implement stick diagrams of CMOS logic gates and observe analog simulation outputs.

Features of MICROWIND2

- CMOS layout and stick diagram design
- Physical-level transistor visualization
- Analog simulation and waveform analysis
- Design rule-based layout implementation
- Visualization of metal, diffusion, and polysilicon layers

Lab 1 CMOS Logic Gate in DSCH2

1.1 Inverter / NOT Gate

A NOT gate, also known as an inverter, produces the complement of its input.

Schematic Image

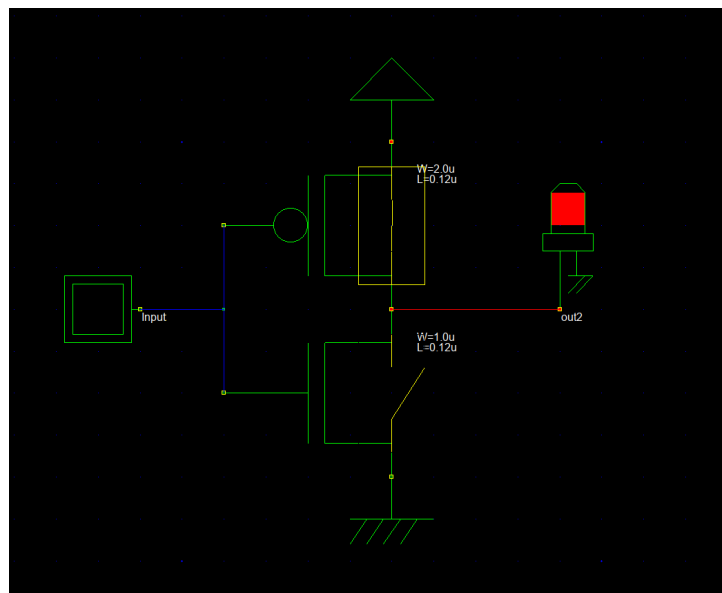


Figure 1.1: Schematic of the 1-input Inverter

Truth Table

A	Y
0	1
1	0

Observation

From the truth table and simulation results, it is observed that the output of the inverter is always the logical complement of the input. When the input $A = 0$, the output Y becomes logic HIGH (1), and when the input $A = 1$, the output Y becomes logic LOW (0). This confirms that the inverter correctly performs logical inversion for all possible input conditions.

1.2 2-input NAND

A 2-input NAND gate is a universal logic gate that produces a logic LOW output only when both inputs are logic HIGH. For all other input combinations, the output remains logic HIGH. Due to its universality, complex digital circuits can be implemented using only NAND gates.

Schematic Image

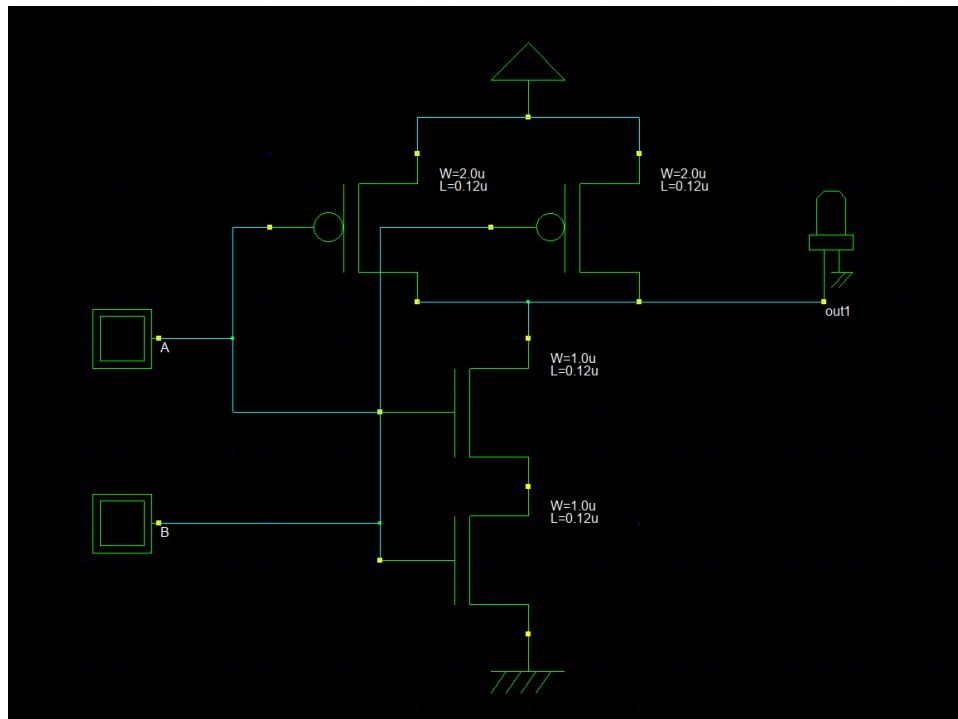


Figure 1.2: Schematic of the 2-input NAND

Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Observation

From the simulation results in DSCH2, it was observed that the output of the 2-input NAND gate remains at logic HIGH for all input combinations except when both inputs A and B are logic HIGH. When either input is logic LOW, the output stays HIGH. The output transitions to logic LOW only for the input condition A = 1 and B = 1. These observations are fully consistent with the theoretical truth table of a 2-input NAND gate, confirming correct logical operation of the designed schematic.

1.3 3-input NAND

A 3-input NAND gate outputs a logic LOW only when all three inputs are logic HIGH. For any other input combination, the output remains logic HIGH.

Schematic Image

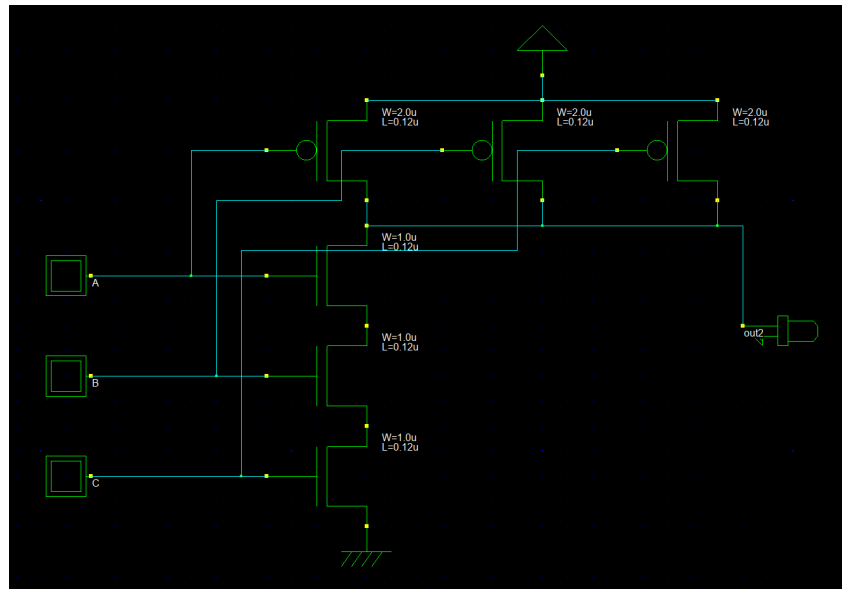


Figure 1.3: Schematic of the 3-input NAND

Truth Table

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Observation

From the truth table and simulation results, it was observed that the output of the 3-input NAND gate remains at logic HIGH for all input combinations except when all three inputs (A, B, and C) are simultaneously at logic HIGH. In the case where $A = 1$, $B = 1$, and $C = 1$, the output transitions to logic LOW. This behavior confirms that the gate performs the logical negation of the AND operation for three inputs. The observed output values from the DSCH2 simulation precisely matched the theoretical truth table, indicating correct functionality of the 3-input NAND gate.

1.4 2-input NOR

A 2-input NOR gate produces a logic HIGH output only when both inputs are logic LOW. For all other input combinations, the output is logic LOW. NOR gates are also universal gates in digital logic design.

Schematic Image

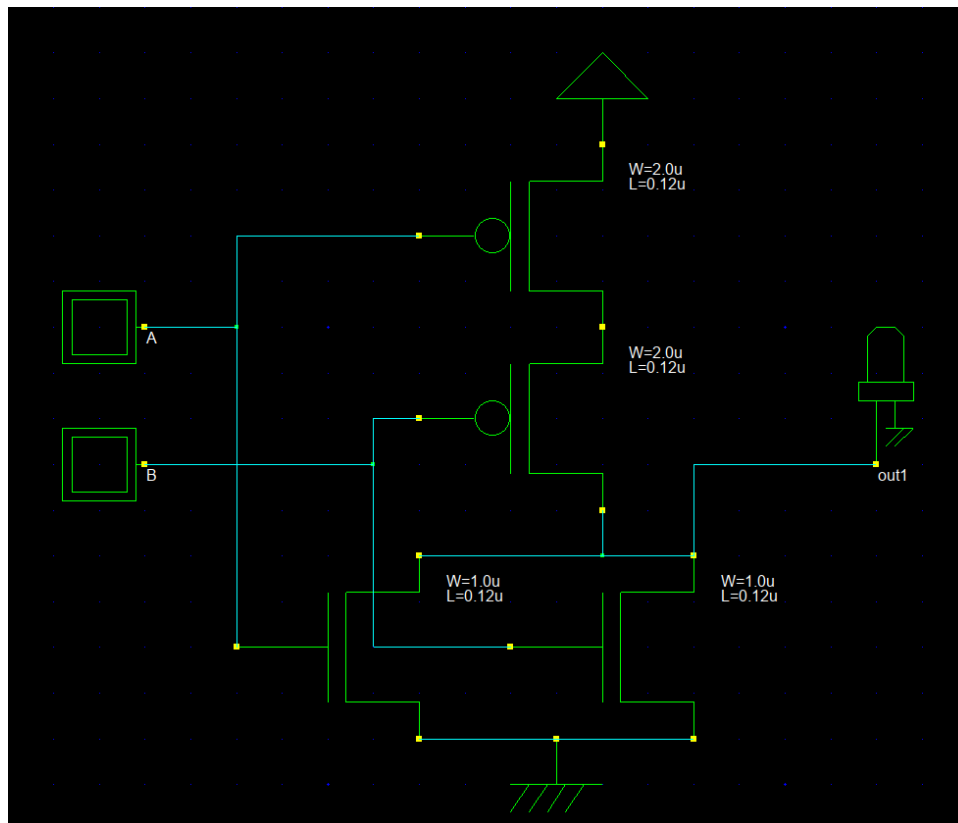


Figure 1.4: Schematic of the 2-input NOR

Truth Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Observation

From the truth table and simulation results, it is observed that the output of the 2-input NOR gate remains at logic HIGH only when both inputs A and B are at logic LOW. For all other input combinations where at least one input is logic HIGH, the output transitions

to logic LOW. This behavior confirms that the circuit performs the logical NOR operation correctly, as verified through DSCH2 simulation.

1.5 3-input NOR

Schematic Image

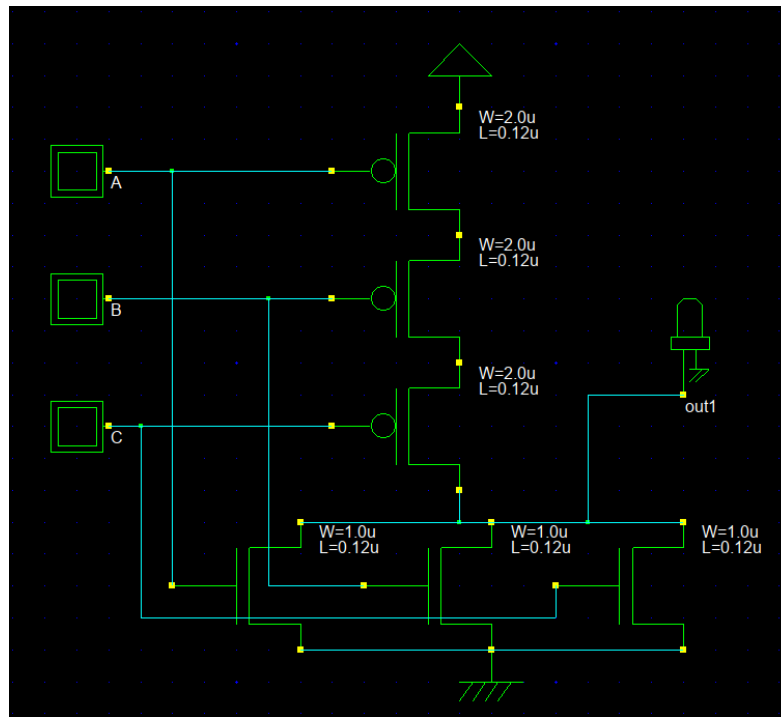


Figure 1.5: Schematic of the 3-input NOR

Truth Table

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Observation

From the simulation results, it is observed that the output of the 3-input NOR gate becomes logic HIGH only when all three inputs (A, B, and C) are at logic LOW. For any input combination where at least one input is HIGH, the output remains logic LOW. This behavior is

consistent with the truth table of the 3-input NOR gate. The DSCH2 simulation output accurately reflects the expected NOR operation, confirming the correctness of the schematic design.

1.6 AND Gate

An AND gate outputs logic HIGH only when all inputs are HIGH.

Schematic Image

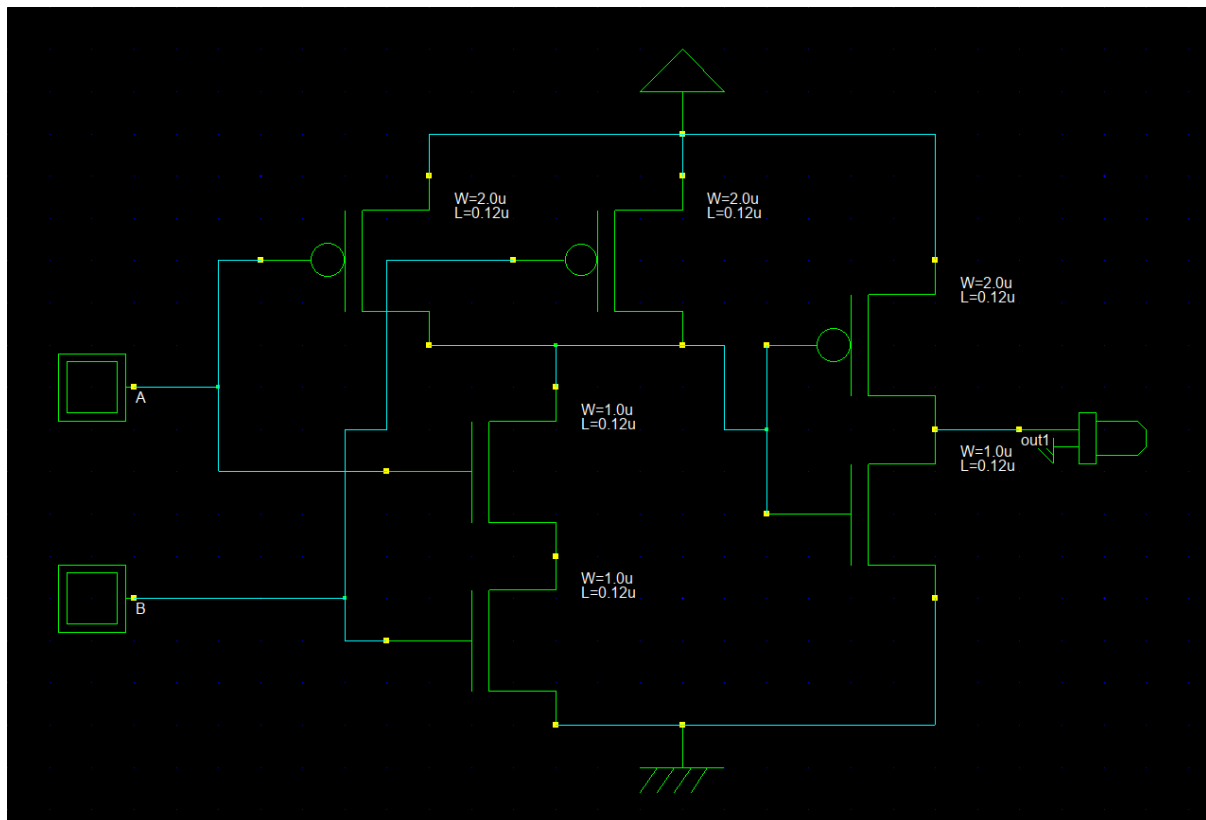


Figure 1.6: Schematic of the AND Gate

Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Observation

From the simulation results of the 2-input AND gate in DSCH2, it was observed that the output remains at logic LOW when any one or both of the inputs are at logic LOW. The

output transitions to logic HIGH only when both inputs A and B are simultaneously at logic HIGH. This behavior is consistent with the truth table of an AND gate, confirming that the implemented schematic operates correctly.

1.7 OR Gate

An OR gate produces logic HIGH when at least one input is HIGH.

Schematic Image

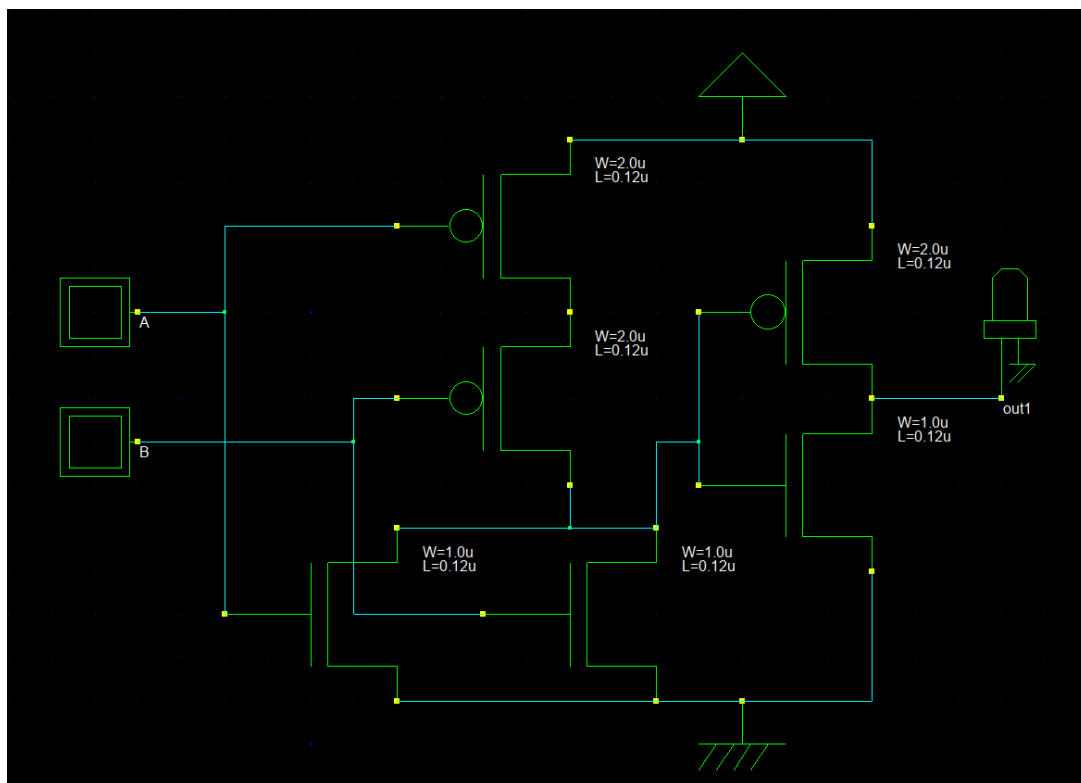


Figure 1.7: Schematic of the OR Gate

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Observation

From the simulation results obtained in DSCH2, it was observed that the output of the 2-input OR gate becomes logic HIGH whenever at least one of the inputs is at logic HIGH.

When both inputs A and B are at logic LOW, the output remains logic LOW. The simulation output exactly follows the theoretical truth table of the OR gate, confirming that the gate correctly performs the logical OR operation.

1.8 XOR Gate

An XOR gate outputs logic HIGH when the inputs are different.

Schematic Image

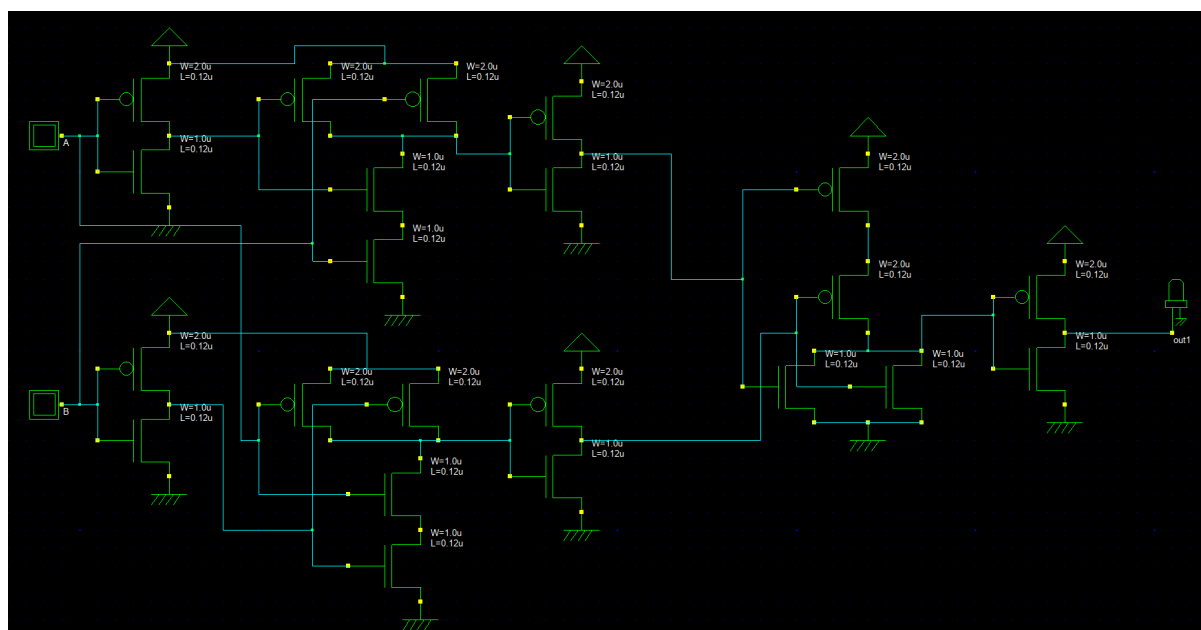


Figure 1.8: Schematic of the XOR Gate

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Observation

From the simulation results of the 2-input XOR gate in DSCH2, it was observed that the output remains logic LOW when both inputs are at the same logic level, i.e., when both inputs are LOW (0,0) or both inputs are HIGH (1,1). The output becomes logic HIGH when the inputs are different, i.e., for input combinations (0,1) and (1,0). These observations exactly match the theoretical truth table of the XOR gate, confirming that the circuit correctly performs the exclusive-OR operation.

Lab 2 Implementing Sequential Circuit

Introduction

The objective of this laboratory experiment is to implement given Boolean expressions using CMOS logic and verify their functional correctness through schematic design and timing simulation. The circuits were designed using the DSCH2 schematic editor, following standard CMOS pull-up and pull-down network principles. Truth tables and timing diagrams were analyzed to validate logical behavior.

Objective

The objectives of this VLSI-Lab experiment are as follows:

- To implement given Boolean expressions using CMOS logic design methodology.
- To design schematic-level CMOS circuits using the DSCH2 tool.
- To verify the logical correctness of the implemented circuits using truth tables.
- To analyze timing diagrams to observe signal transitions and propagation behavior.
- To develop a clear understanding of CMOS pull-up and pull-down network operation.

2.1 Function 1: $(ABC + D)'$

Boolean Function

$$Y = (ABC + D)'$$

Truth Table

A	B	C	D	Y
0	0	0	0	1
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	1	1	0	0
X	X	X	1	0

Schematic Image

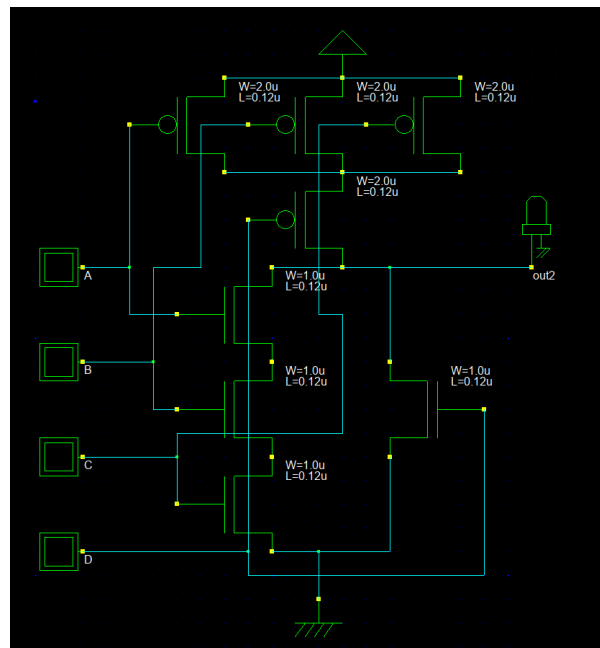


Figure 2.1: CMOS Schematic of $(ABC + D)'$

Timing Diagram

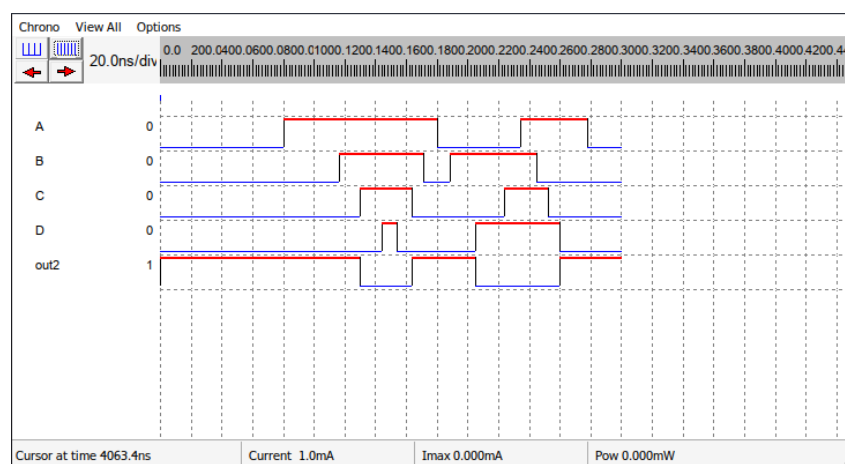


Figure 2.2: Timing Diagram of $(ABC + D)'$

The timing diagram shows that the output remains HIGH except when all inputs A, B, and C are HIGH or when input D is HIGH.

Observation

The simulation confirms that the output goes LOW when either $ABC = 1$ or $D = 1$. This matches the expected inverted logic behavior.

2.2 Function 2: $((AB + CD)E)'$

Boolean Function

$$Y = ((AB + CD)E)'$$

Truth Table

A	B	C	D	E	Y
0	X	X	X	X	1
1	1	X	X	1	0
0	X	1	1	1	0
1	X	X	X	1	1

Schematic Image

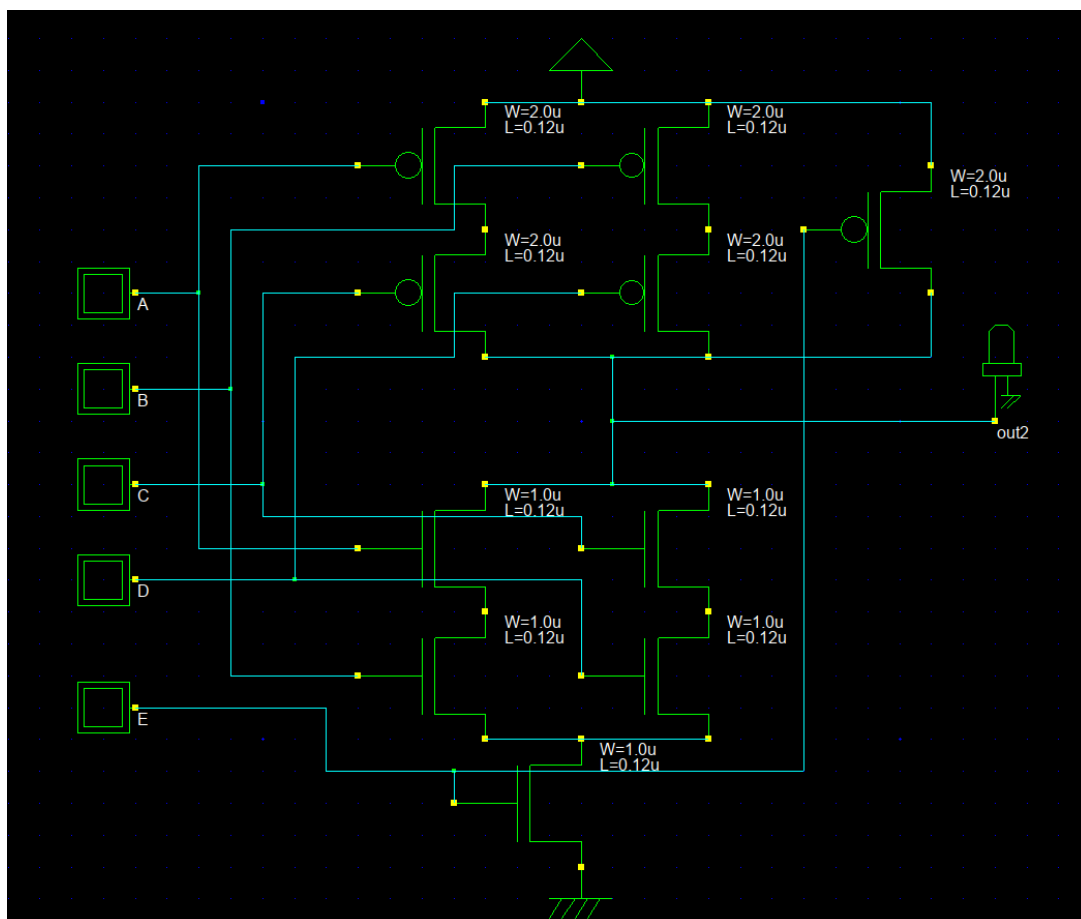


Figure 2.3: CMOS Schematic of $((AB + CD)E)'$

Timing Diagram

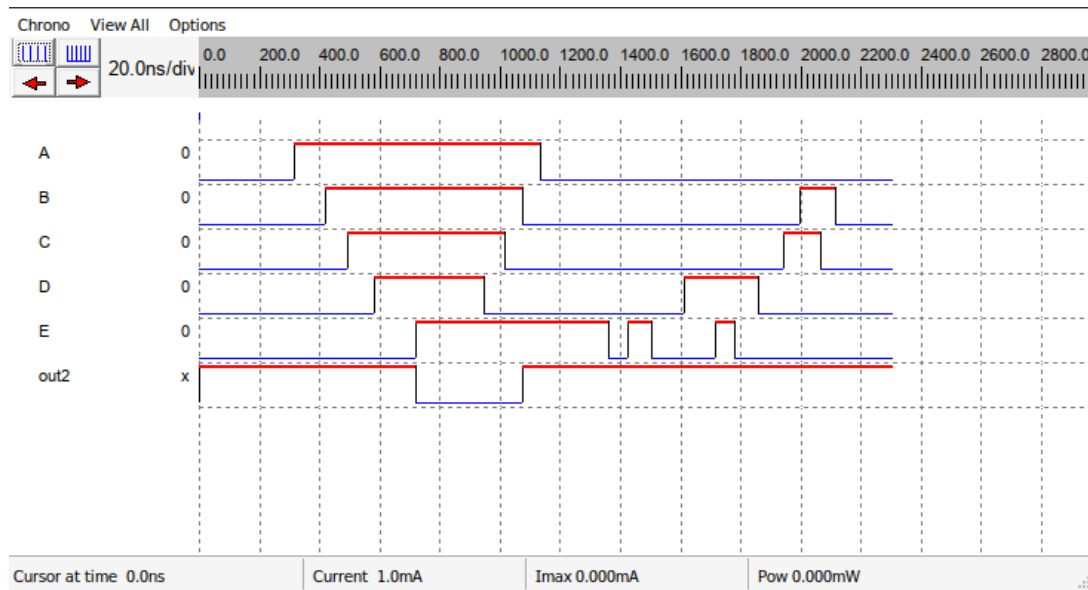


Figure 2.4: Timing Diagram of $((AB + CD)E)'$

The output transitions LOW only when E is HIGH and either AB or CD is satisfied.

Observation

The circuit output follows the complemented logic, validating correct CMOS pull-up and pull-down network operation.

2.3 Function 3: $((AB + C)(A + B))'$

Boolean Function

$$Y = ((AB + C)(A + B))'$$

Truth Table

A	B	C	Y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	0	0
1	0	1	0
1	1	1	0

Schematic Image

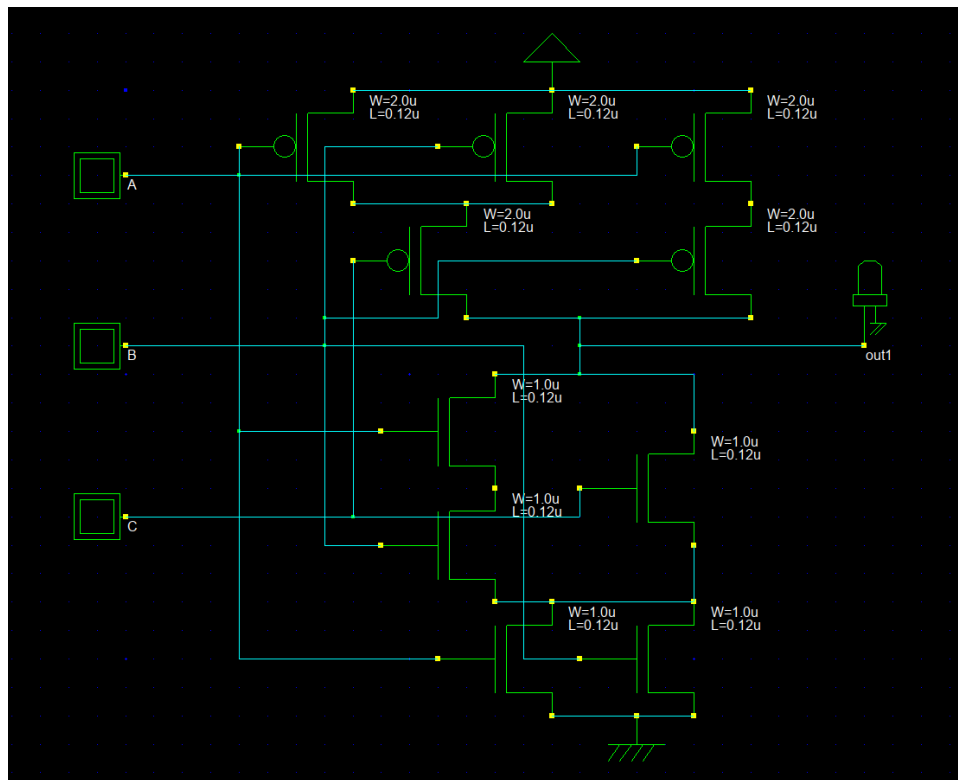


Figure 2.5: CMOS Schematic of $((AB + C)(A + B))'$

Timing Diagram

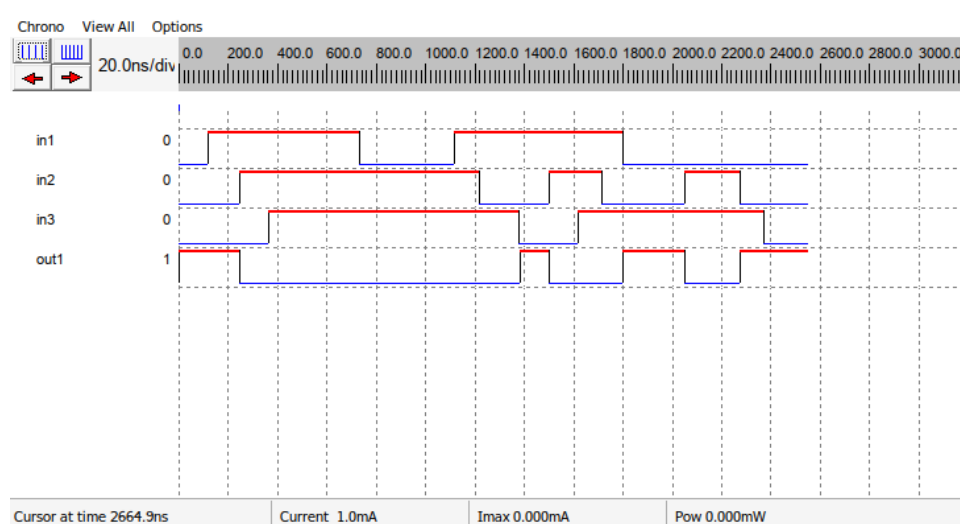


Figure 2.6: Timing Diagram of $((AB + C)(A + B))'$

The output remains HIGH only when the AND of the two expressions is false.

Observation

Simulation results matched the inverted product-of-sums logic expression.

2.4 Function 4: $(A + B + C)D$

Boolean Function

$$Y = (A + B + C)D$$

Truth Table

A	B	C	D	Y
X	X	X	0	0
1	X	X	1	1
0	1	X	1	1
0	0	1	1	1
0	0	0	1	0

Schematic Image

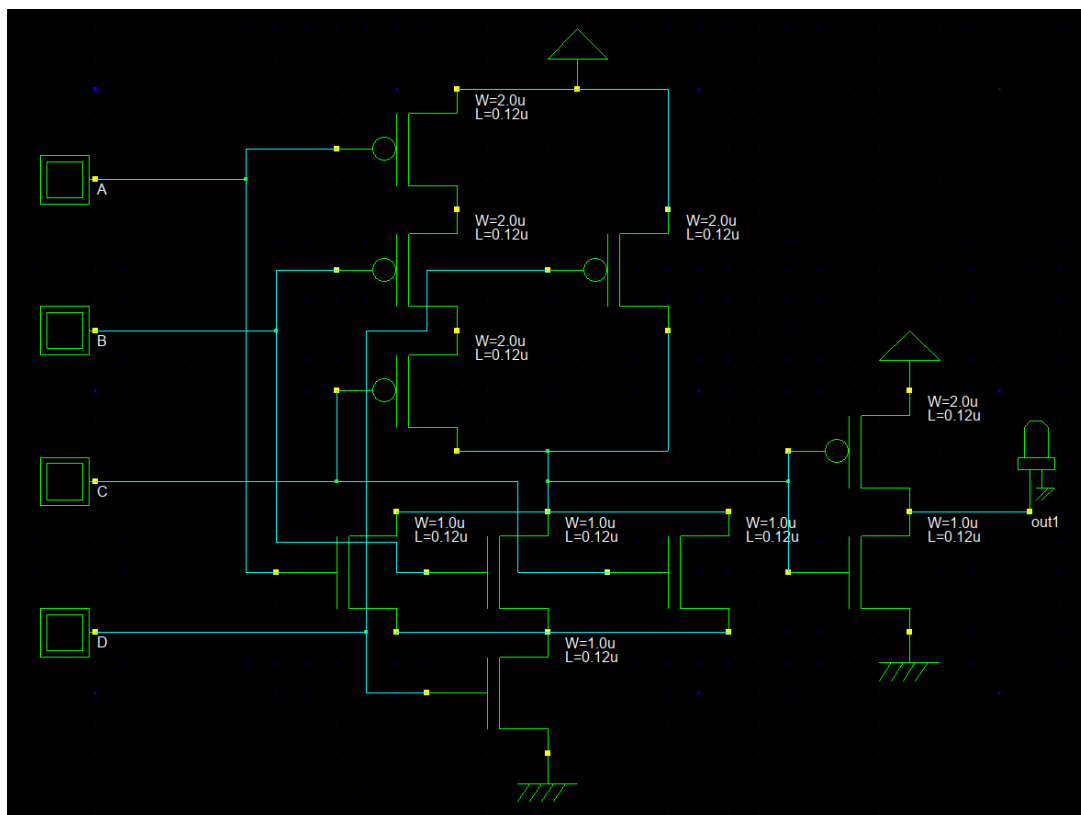


Figure 2.7: CMOS Schematic of $(A + B + C)D$

Timing Diagram

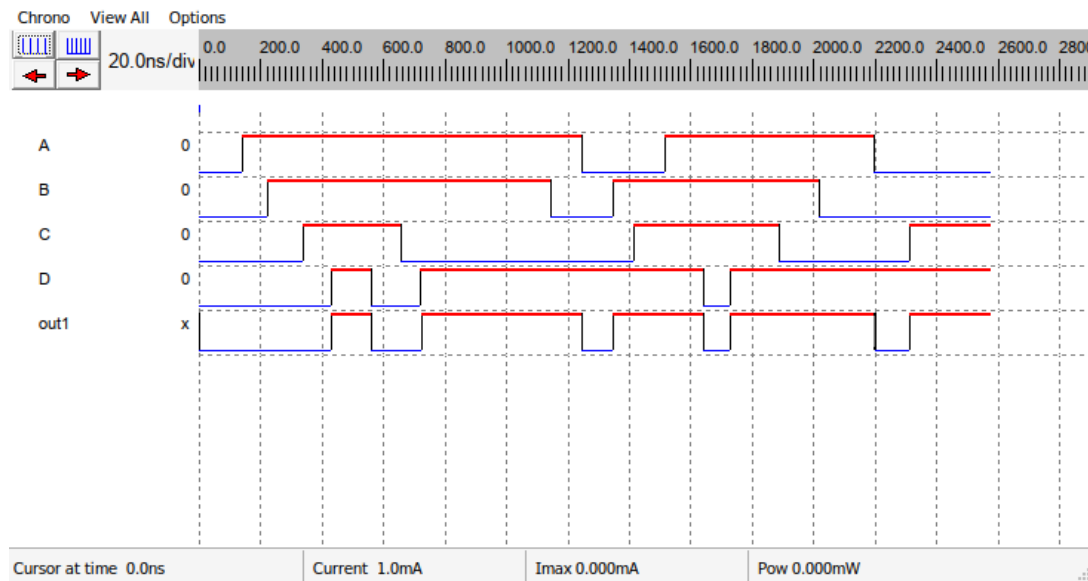


Figure 2.8: Timing Diagram of $(A + B + C)D$

The output is HIGH only when D is HIGH and at least one of A, B, or C is HIGH.

Observation

The simulation confirms correct AND–OR logic behavior.

2.5 Function 5: $((A + B)CD)'$

Boolean Function

$$Y = ((A + B)CD)'$$

Truth Table

A	B	C	D	Y
X	X	X	0	1
0	0	1	1	1
1	X	1	1	0
X	1	1	1	0

Schematic Image

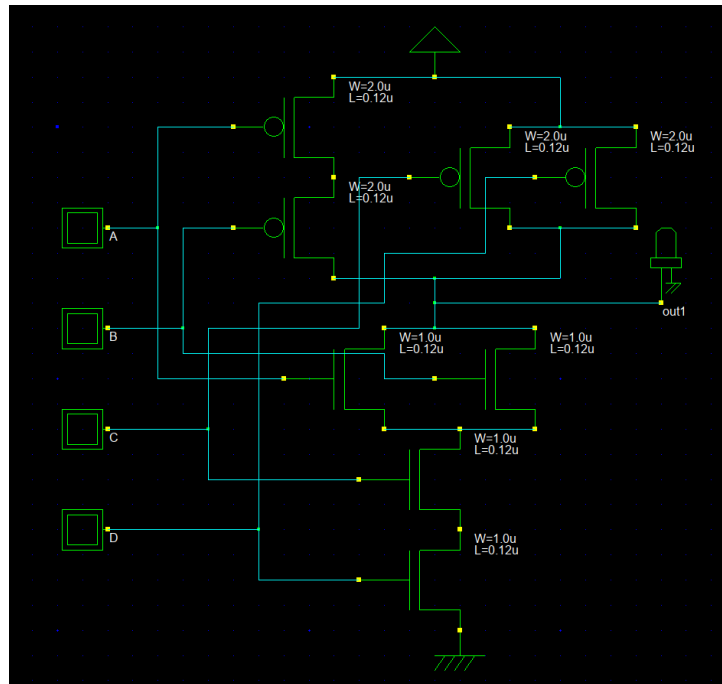


Figure 2.9: CMOS Schematic of $((A + B)CD)'$

Timing Diagram

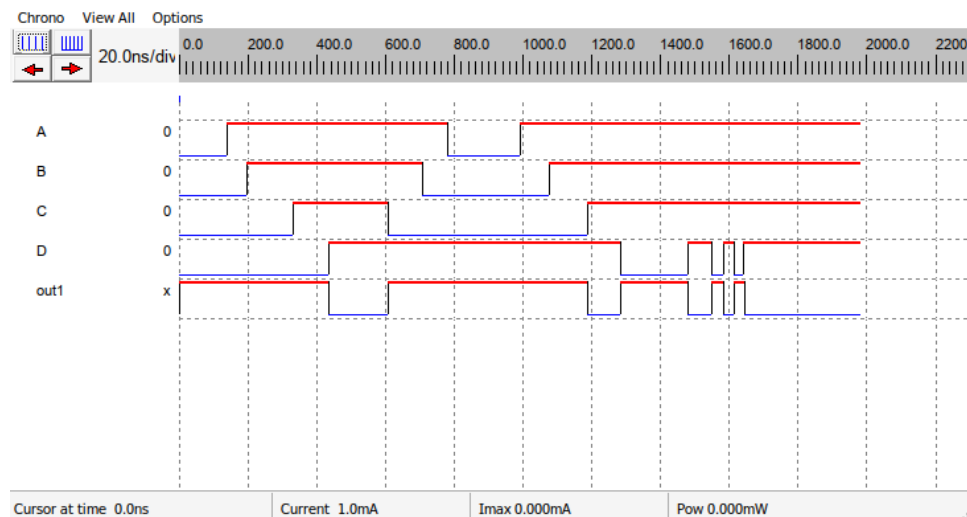


Figure 2.10: Timing Diagram of $((A + B)CD)'$

Observation

The output goes LOW only when C and D are HIGH and either A or B is HIGH. Simulation results confirm correct inverted AND-OR CMOS implementation.

Lab 3 Adder Circuits

Introduction

A combinational logic circuit which is designed to add two binary digits is called as a half adder. The half adder provides the output along with a carry value (if any). The half adder circuit is designed by connecting an EX-OR gate and one AND gate. It has two input terminals and two output terminals for sum and carry.

Objective

The objectives of this VLSI-Lab experiment are as follows:

- To implement Adder Circuit using CMOS logic design methodology.
- To design schematic-level Adder circuits using the DSCH2 tool.
- To verify the logical correctness of the implemented circuits using truth tables.
- To analyze timing diagrams to observe signal transitions and propagation behavior.
- To develop a clear understanding of Adder Circuit.

3.1 Half Adder Function

Boolean Function

$$Sum = AB' + A'B$$

$$Carry = A B$$

Truth Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Schematic Image

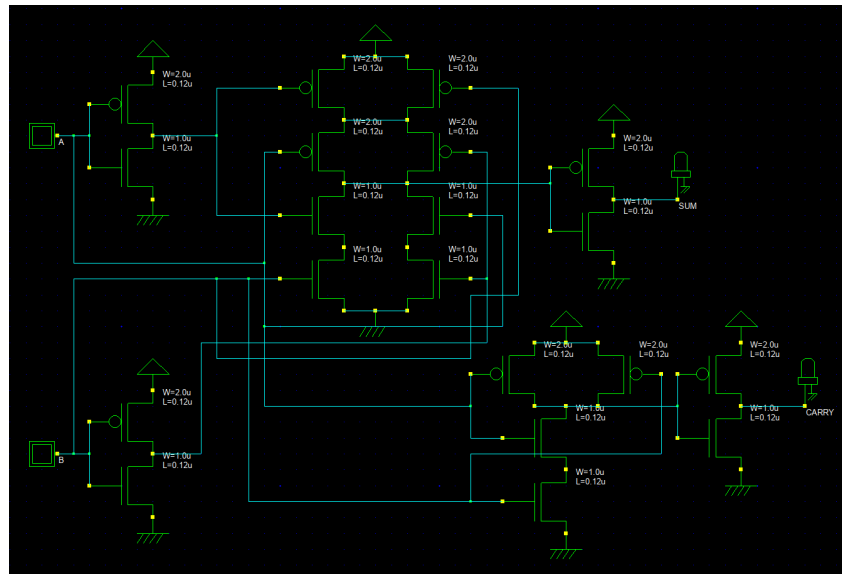


Figure 3.1: CMOS Schematic of Half-Adder

Timing Diagram

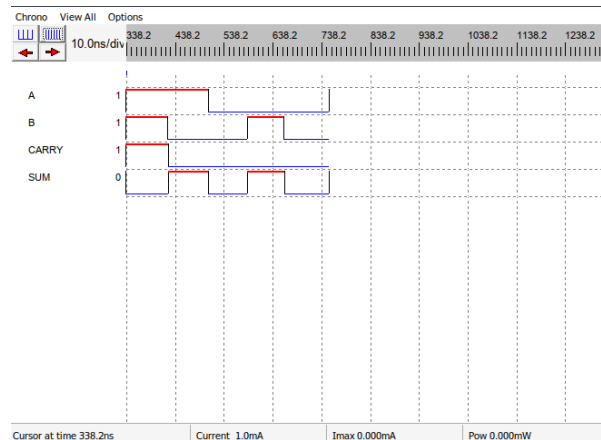


Figure 3.2: Timing Diagram of Half-Adder

Observation

The timing diagram of a half adder illustrates the sequence of operations that occur when two binary inputs are added. The diagram typically shows the timing of the XOR operation for the sum and the timing of the AND operation for the carry. The timing diagram can help in understanding the propagation delay and the timing constraints of the circuit. It is important to note that the timing diagram should be adjusted to account for the specific timing requirements of the circuit design, such as rise/fall time and maximum input frequency.

Lab 4 Flip-Flop Circuits

Introduction

This laboratory experiment focuses on the design and implementation of a flip-flop circuit using CMOS logic in the DSCH2 environment. Flip-flops are fundamental sequential elements used for data storage and synchronization in digital systems. In this experiment, a D Flip-Flop was designed using CMOS-based logic gates, and its functionality was verified through truth table analysis and timing simulation. The behavior of the circuit was observed with respect to clock-triggered input transitions.

Objective

The objectives of this laboratory experiment are as follows:

- To design and implement a flip-flop circuit using CMOS logic.
- To develop schematic-level circuits using the DSCH2 tool.
- To derive and verify the truth table of the flip-flop from simulation.
- To analyze timing diagrams to observe clock-triggered behavior.
- To understand the operation of sequential circuits and data storage elements.

4.1 Function of FlipFlop Circuit

The D Flip-Flop is a clock-controlled bistable device that transfers the input data (D) to the output (Q) on the active edge of the clock signal.

$$Q_{next} = D$$

The output changes only at the triggering edge of the clock, ensuring synchronized data storage.

Truth Table

Clock	D	Q(next)
↑	0	0
↑	1	1
No Edge	X	Q(previous)

Schematic Image

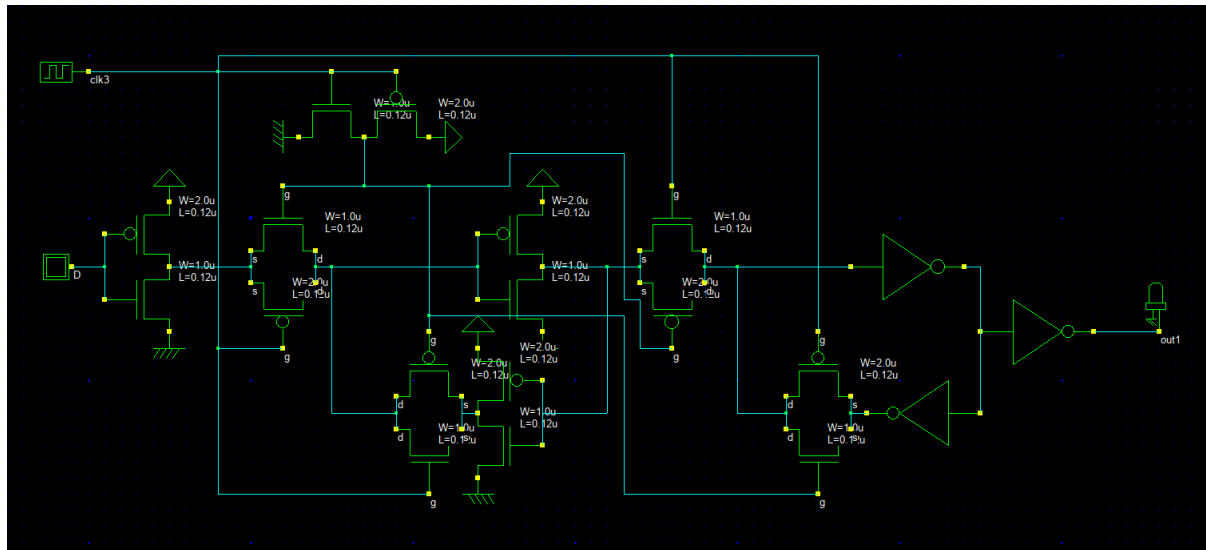


Figure 4.1: CMOS Schematic of D Flip-Flop in DSCH2

Timing Diagram

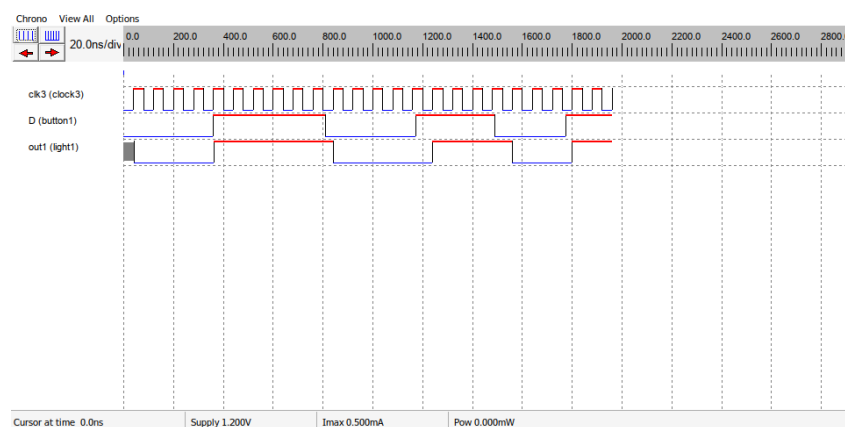


Figure 4.2: Timing Diagram of D Flip-Flop

Observation

The timing diagram shows that the output Q follows the input D only at the rising edge of the clock signal. Between clock edges, the output remains constant regardless of changes in the input.

From the simulation results, it was observed that the D Flip-Flop changes its output state only at the rising edge of the clock signal. When the clock transitions from LOW to HIGH, the output Q takes the value of the input D. At all other times, the output remains unchanged, preserving the previously stored value. This behavior confirms the correct operation of the edge-triggered D Flip-Flop implemented using CMOS logic.

Lab 5 Stick Diagram in MICROWIND2

Introduction

In VLSI design, the transition from schematic-level representation to physical layout is a critical step in the design process. Stick diagrams serve as an intermediate abstraction that simplifies the visualization of circuit layouts without considering exact dimensions. These diagrams use standardized color codes to represent different layers such as polysilicon, diffusion regions, and metal interconnections.

Microwind2 is a layout design tool that allows designers to implement and simulate CMOS circuits at the physical level. In this experiment, stick diagrams of basic logic gates were implemented to understand layout organization, transistor placement, and interconnection strategies. The study of stick diagrams helps in developing an understanding of design rules, layout optimization, and area-efficient circuit implementation in VLSI systems.

Objective

The objectives of this laboratory experiment are as follows:

- To understand the concept and importance of stick diagrams in VLSI layout design.
- To implement stick diagrams of basic CMOS logic gates using Microwind2.
- To represent transistor-level circuits using simplified, technology-independent layouts.
- To analyze the arrangement of polysilicon, diffusion, and metal layers in CMOS design.
- To develop a clear understanding of the relationship between schematic design and physical layout.

5.1 1 Input Inverter (NOT Gate)

The inverter, also known as a NOT gate, is the most fundamental logic gate in digital design. It produces the logical complement of the input signal. In CMOS technology, the inverter is implemented using one PMOS transistor in the pull-up network and one NMOS transistor in the pull-down network. The stick diagram represents this structure using polysilicon for gate control, diffusion regions for source and drain, and metal layers for interconnections. The inverter serves as the basic building block for more complex CMOS circuits.

Stick Diagram

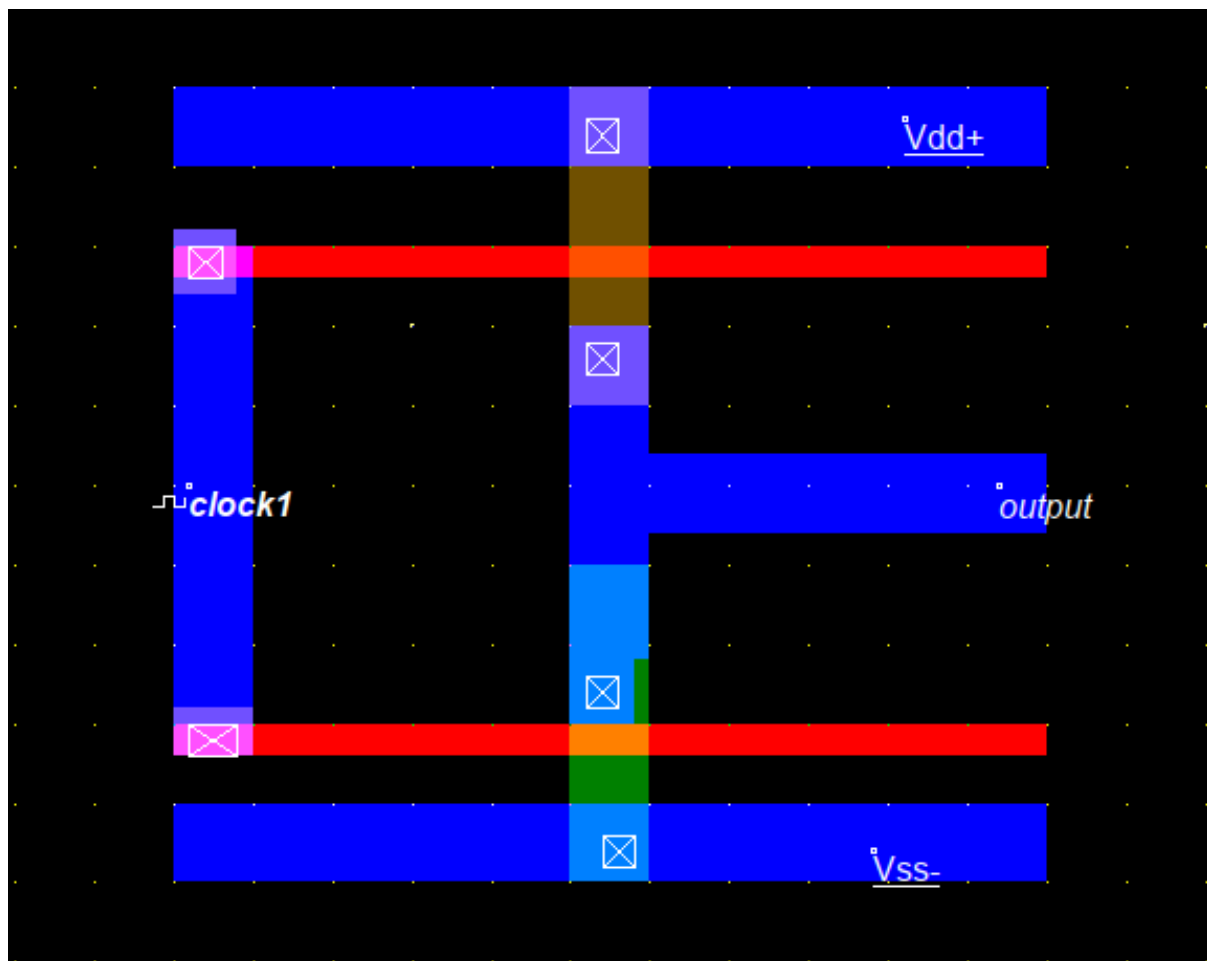


Figure 5.1: Stick Diagram of Inverter (NOT Gate) in MICROWIND2

Output

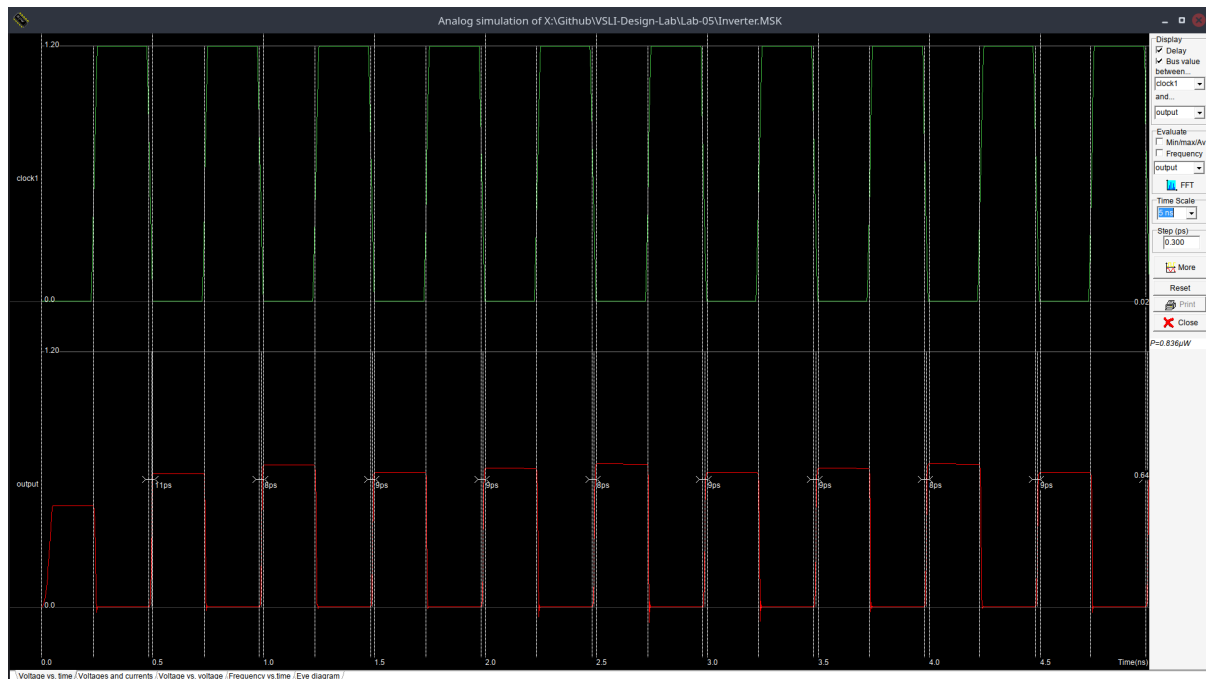


Figure 5.2: Analog Simulation of Inverter (NOT Gate)

Observation

From the output waveform, it is observed that the inverter produces the exact complement of the input signal. When the input is at logic HIGH, the NMOS transistor conducts and pulls the output to logic LOW. Conversely, when the input is at logic LOW, the PMOS transistor conducts and pulls the output to logic HIGH. The transition between output states occurs immediately following the input change, demonstrating correct switching behavior. The observed output confirms the proper operation of the CMOS inverter as expected from its theoretical truth table.

5.2 2 Input NAND Gate

A 2-input NAND gate is a fundamental universal logic gate that produces a logic LOW output only when both inputs are logic HIGH. In CMOS layout design, the NAND gate is implemented using two NMOS transistors connected in series in the pull-down network and two PMOS transistors connected in parallel in the pull-up network. The stick diagram represents this structure using simplified layer representations such as polysilicon for gates, diffusion for active regions, and metal for interconnections.

Stick Diagram

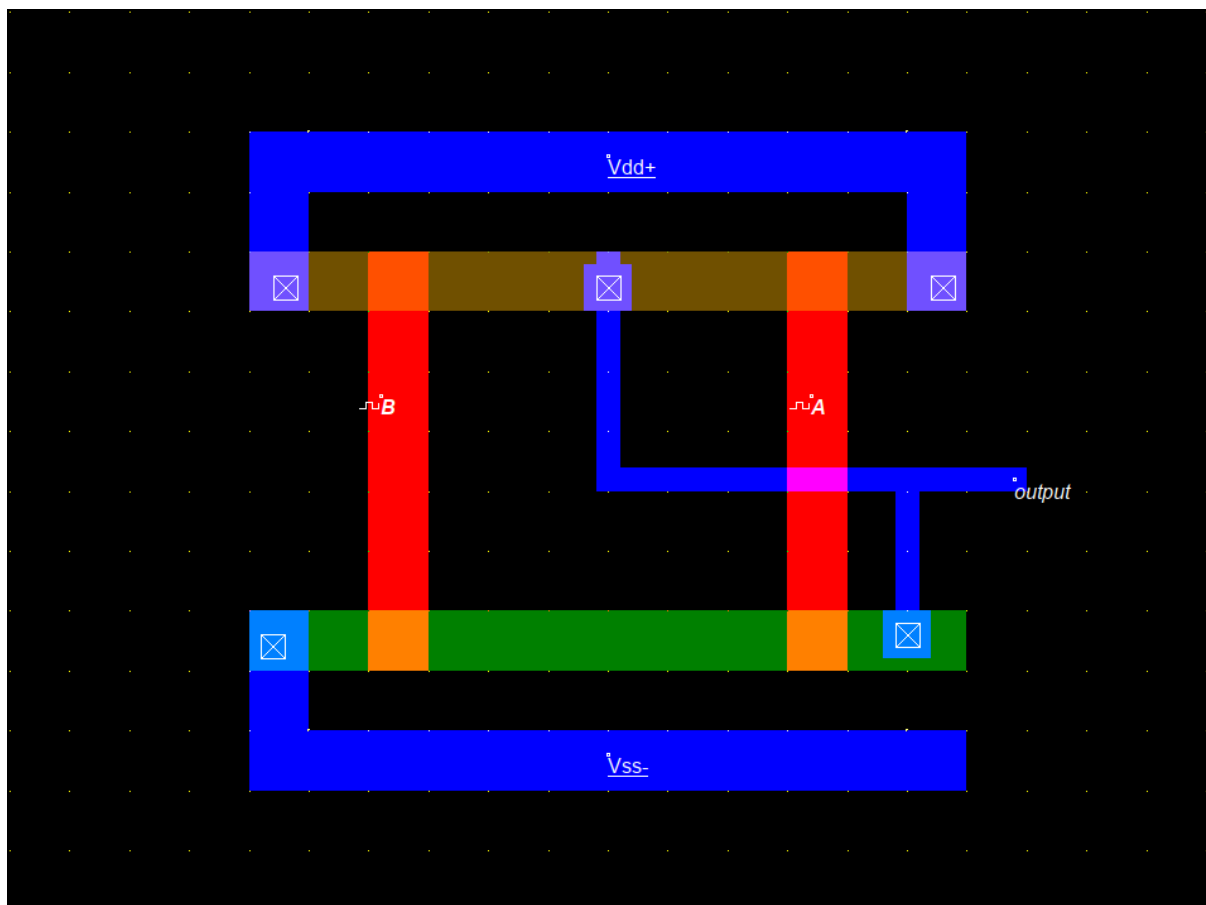


Figure 5.3: Stick Diagram of 2in-NAND Gate in MICROWIND2

Output

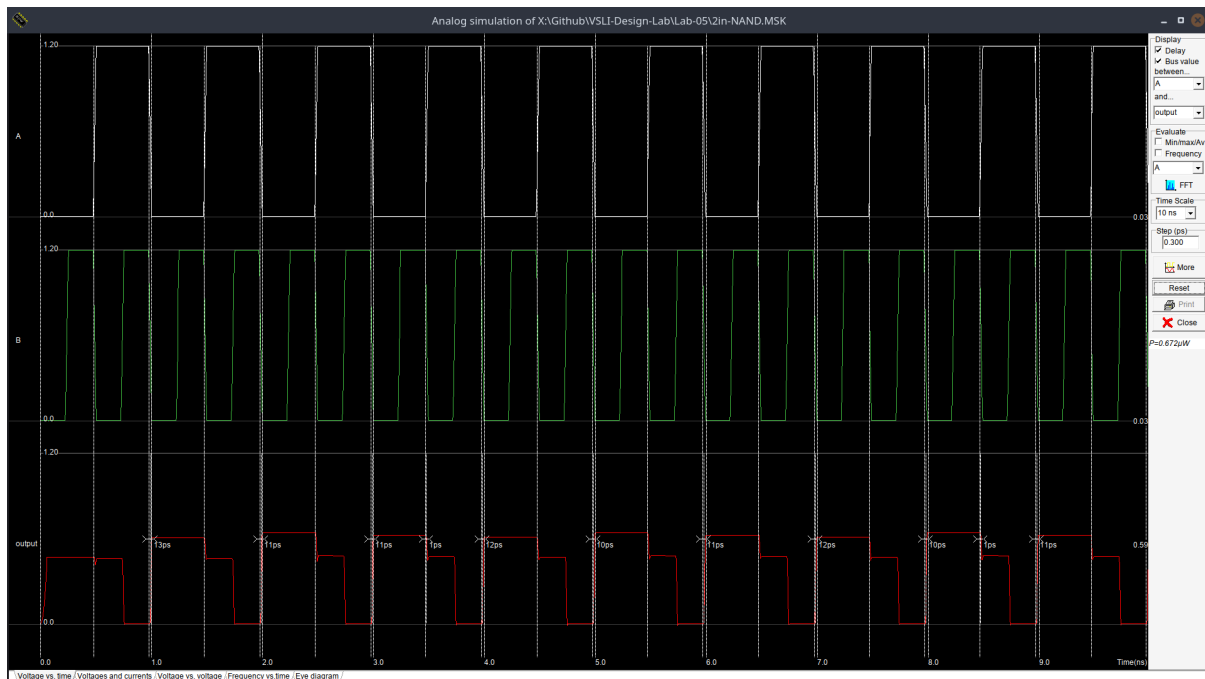


Figure 5.4: Analog Simulation of 2in-NAND Gate

Observation

From the simulation results obtained in Microwind2, it is observed that the output remains at logic HIGH for all input combinations except when both inputs are HIGH. When inputs A and B are simultaneously at logic HIGH, the series NMOS transistors in the pull-down network conduct, creating a path to ground and forcing the output to logic LOW. For all other input conditions, at least one PMOS transistor in the pull-up network remains active, maintaining the output at logic HIGH. The observed output waveform matches the expected truth table of a 2-input NAND gate, confirming the correctness of the stick diagram implementation.

5.3 2 Input NOR Gate

The 2-input NOR gate is a fundamental digital logic gate that produces a logic HIGH output only when both inputs are at logic LOW. In CMOS implementation, the NOR gate is realized using parallel NMOS transistors in the pull-down network and series PMOS transistors in the pull-up network. In the stick diagram representation using Microwind2, different layers such as polysilicon, diffusion, and metal are arranged according to standard layout conventions to form the required transistor connections. The stick diagram provides a simplified visualization of the physical layout without considering exact dimensions.

Stick Diagram

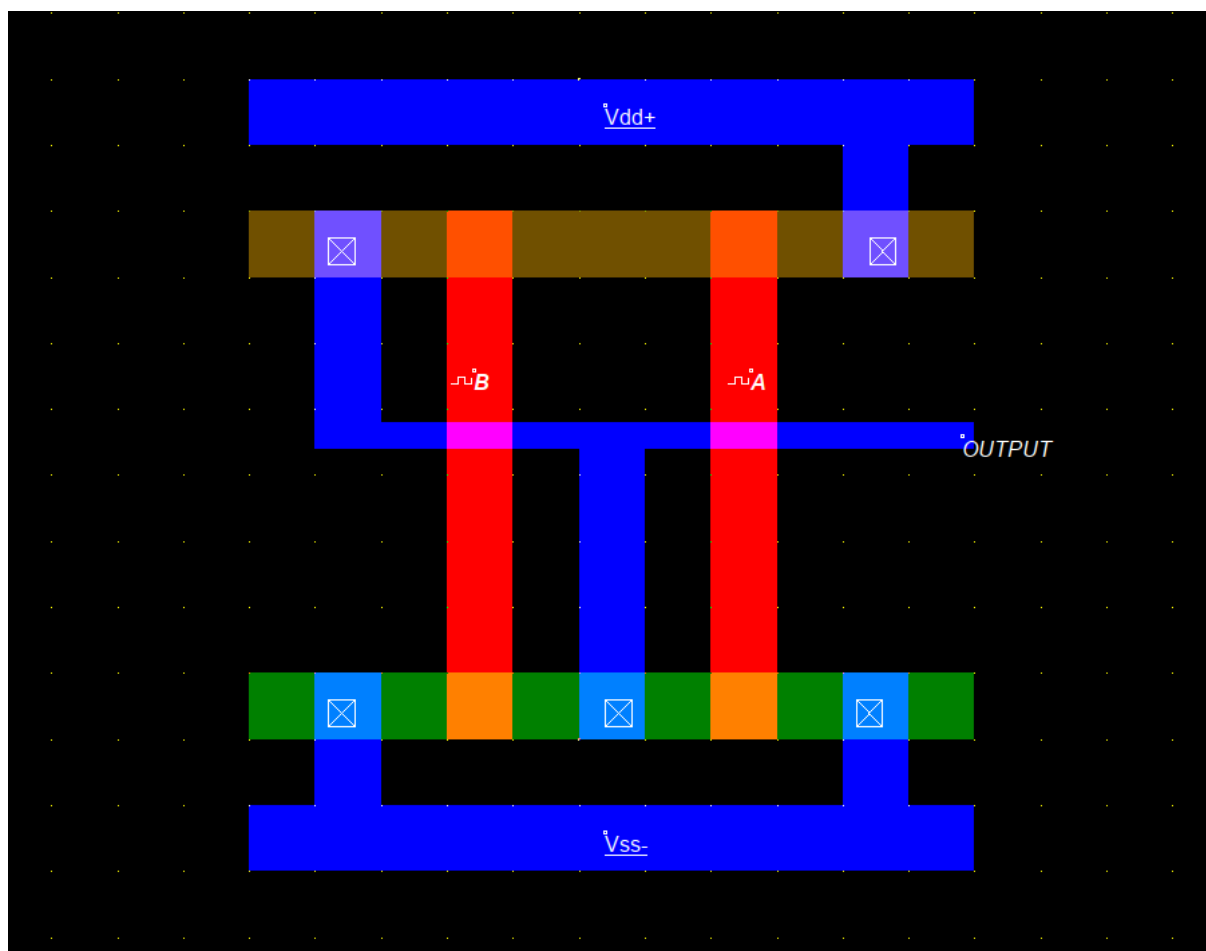


Figure 5.5: Stick Diagram of 2in-NOR Gate in MICROWIND2

Output

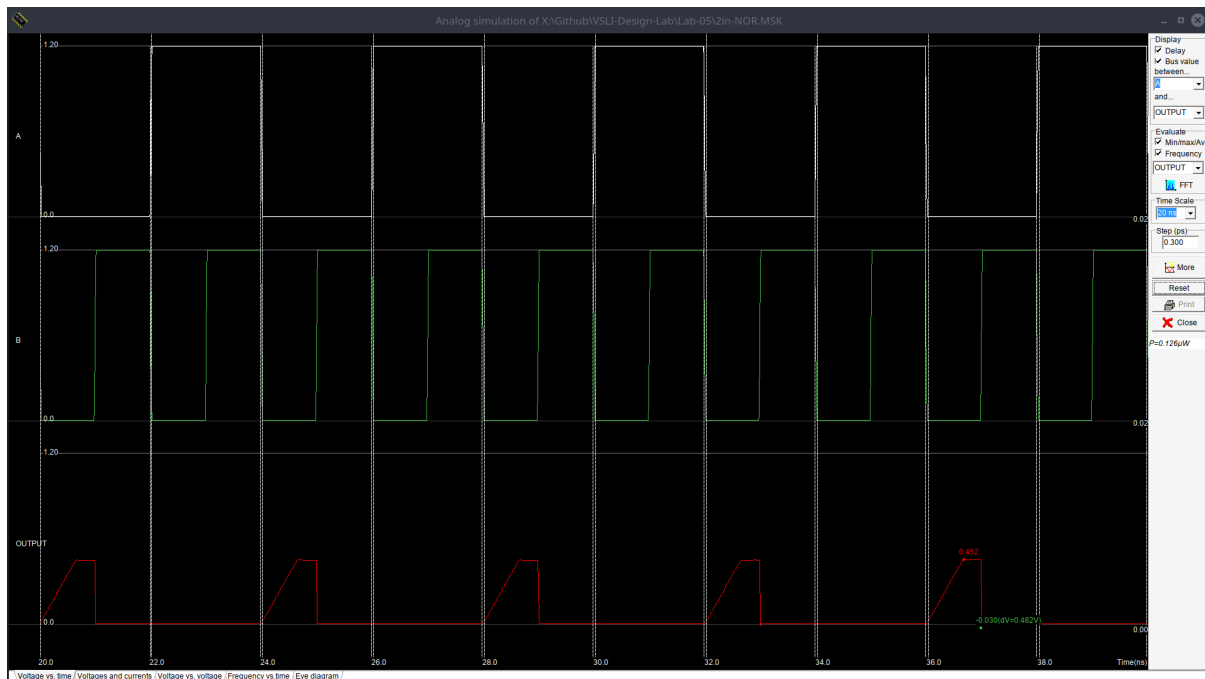


Figure 5.6: Analog Simulation of 2in-NOR Gate

Observation

From the simulation results, it is observed that the output of the 2-input NOR gate remains logic HIGH only when both inputs are LOW (0,0). For all other input combinations, i.e., when any one or both inputs are HIGH, the output becomes logic LOW. The output waveform clearly reflects this behavior, showing a HIGH level only in the absence of input signals. This verifies the correct implementation of the NOR logic function in the stick diagram using Microwind2.

Lab 6 Stick Diagram of XOR function

Introduction

In VLSI design, the transformation of logical circuits into physical layouts is an essential step in the design process. Stick diagrams serve as a simplified and technology-independent method for representing circuit layouts, allowing designers to visualize transistor placement and interconnections without focusing on exact dimensions.

The XOR gate is an important combinational logic circuit widely used in arithmetic and digital systems. Its implementation in CMOS requires careful arrangement of pull-up and pull-down networks. In this experiment, the XOR gate is designed using its Boolean expression and represented using a stick diagram in Microwind2. The study of this design helps in understanding layout organization, layer interaction, and efficient circuit realization in VLSI systems.

Objective

The objectives of this laboratory experiment are as follows:

- To understand the concept and importance of stick diagrams in VLSI layout design.
- To implement stick diagrams of basic CMOS logic gates using Microwind2.
- To represent transistor-level circuits using simplified, technology-independent layouts.
- To analyze the arrangement of polysilicon, diffusion, and metal layers in CMOS design.
- To develop a clear understanding of the relationship between schematic design and physical layout.

An XOR (Exclusive-OR) gate is a combinational logic gate that produces a logic HIGH output when the inputs are different and a logic LOW output when the inputs are the same. It is commonly used in digital circuits for comparison, parity checking, and arithmetic operations.

Boolean Function of XOR Gate

$$Y = (AB + A'B')'$$

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Stick Diagram

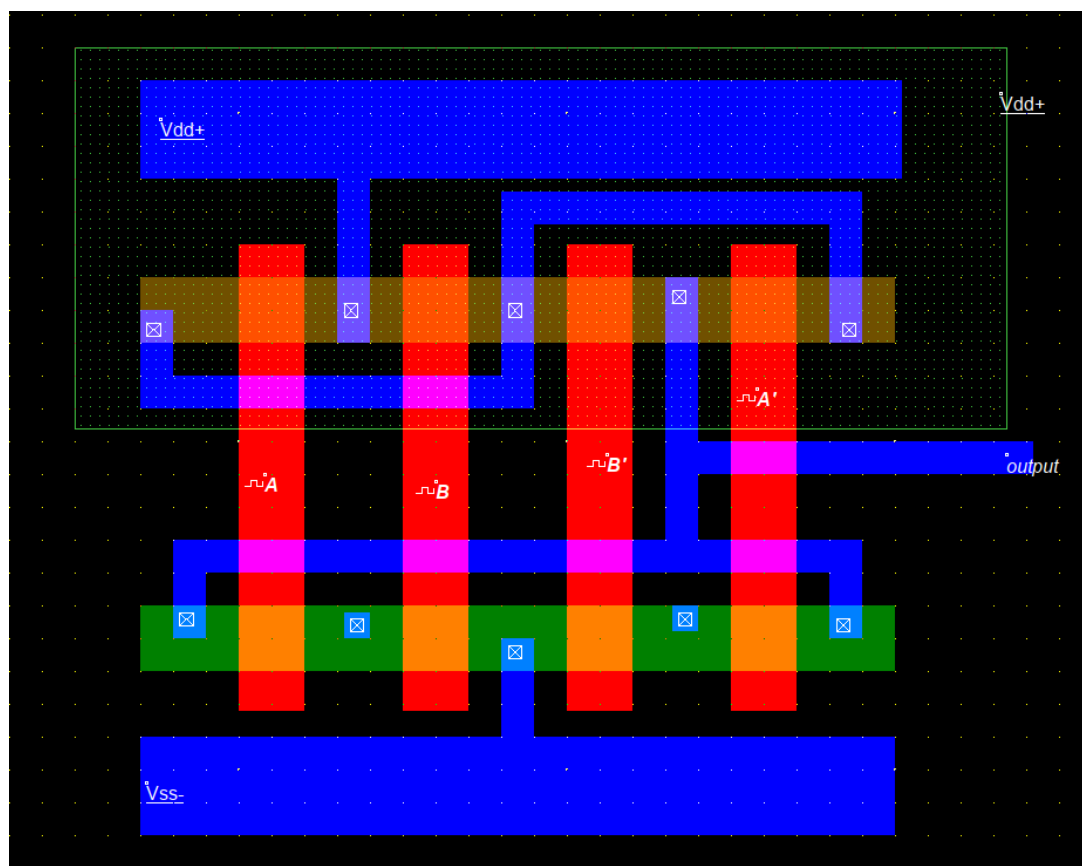


Figure 6.1: Stick Diagram of XOR Gate

Output

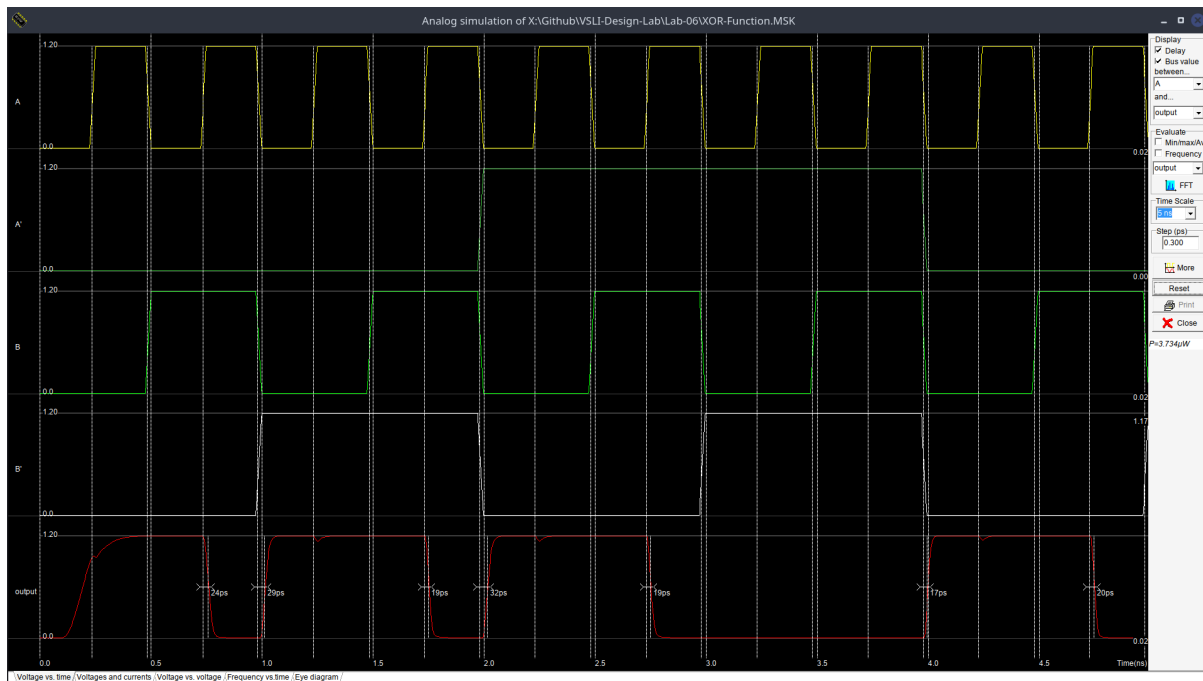


Figure 6.2: Output in Analog Simulation of XOR Gate

Observation

From the implementation and simulation of the XOR gate stick diagram using Microwind2, it was observed that the circuit produces a logic HIGH output when the inputs are different and a logic LOW output when both inputs are the same. The output waveform obtained from the simulation matches the theoretical truth table of the XOR function.

The stick diagram correctly represents the CMOS structure using appropriate placement of polysilicon, diffusion, and metal layers. Proper alignment and interconnection of these layers ensured correct signal flow and circuit operation. The simulation results confirm that the designed layout accurately implements the XOR logic function.

Conclusion

In this VLSI Design Lab course, various CMOS-based digital circuits were successfully designed, implemented, simulated, and analyzed using DSCH2 and MICROWIND2 software tools. The experiments provided practical knowledge of combinational logic circuits, sequential circuits, timing analysis, CMOS implementation techniques, and physical layout representation in VLSI systems.

The implementation of basic logic gates such as NOT, NAND, NOR, AND, OR, and XOR using DSCH2 helped in understanding the operation of CMOS pull-up and pull-down networks. Verification through truth tables and timing simulations confirmed the correct logical behavior of each circuit. The implementation of Boolean expressions further strengthened the understanding of CMOS logic design methodology and transistor level circuit realization.

The design and analysis of Half Adder and D Flip-Flop circuits introduced important concepts of arithmetic operations, sequential logic, data storage, and clock-triggered behavior. Timing diagrams demonstrated signal transitions and propagation characteristics of digital circuits.

The implementation of stick diagrams using MICROWIND2 provided practical experience in translating schematic-level circuits into simplified physical layouts. The arrangement of polysilicon, diffusion, and metal layers helped in understanding transistor placement, routing techniques, and layout organization in CMOS design. Analog simulation outputs verified the correctness of the implemented layouts.

Overall, this laboratory course successfully developed fundamental knowledge and practical skills related to VLSI circuit design, CMOS implementation, timing analysis, and physical layout representation. The experiments provided a strong foundation for advanced study and research in VLSI system design and semiconductor technology.